

Chapter 6D: Adrenal Disease

Cushing Syndrome, Pheochromocytoma, Primary Aldosteronism, Adrenal Insufficiency, and Congenital Adrenal Hyperplasia

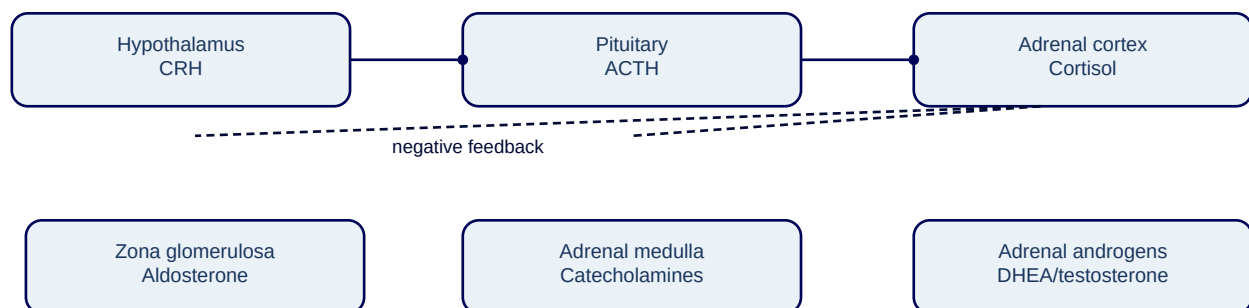
How this chapter was built

This is an original Medvale chapter. Uploaded endocrine pages were used as a coverage map for adrenal disorders, including the source sections on Cushing syndrome, pheochromocytoma, primary hyperaldosteronism, adrenal insufficiency, and congenital adrenal hyperplasia. The diagnostic framing was cross-checked against contemporary endocrine guideline resources.

Why this matters clinically

Adrenal disorders are high-yield because small endocrine organs generate large systemic patterns. Cortisol excess can masquerade as obesity, diabetes, hypertension, depression, osteoporosis, infection risk, and proximal myopathy. Catecholamine excess can present as episodic hypertension, panic-like spells, headache, palpitations, and sweating. Aldosterone excess is a common, treatable cause of secondary hypertension and classically produces hypokalemia and metabolic alkalosis. Cortisol deficiency can be fatal during physiologic stress because patients cannot maintain vascular tone, glucose, sodium, or effective circulating volume. The unifying task is to decide which adrenal hormone is excessive or deficient, determine whether the defect is ACTH/renin driven, then stabilize emergencies before localization.

The uploaded source emphasizes the major adrenal syndromes and their classic testing patterns: Cushing syndrome screening with dexamethasone suppression or urine/salivary cortisol, pheochromocytoma biochemical testing with metanephrines, aldosterone-renin logic in primary aldosteronism, cortisol/ACTH algorithms in adrenal insufficiency, and 21-hydroxylase deficiency in congenital adrenal hyperplasia.



Aldosterone is mainly RAAS/K⁺ controlled; medulla is sympathetic; androgens rise with ACTH-driven shunting.

Core illness scripts

Disorder	Hormonal problem	Clinical pattern	Diagnostic anchor
Cushing syndrome	Too much cortisol from exogenous steroids, pituitary ACTH, adrenal tumor, or ectopic ACTH	Central weight gain, wide purple striae, proximal weakness, easy bruising, diabetes, hypertension, osteoporosis, infection risk	Screen with 1-mg dexamethasone suppression, late-night salivary cortisol, or 24-hour urine free cortisol; then ACTH to classify
Pheochromocytoma/paraganglioma	Catecholamine-secreting chromaffin tumor	Episodic headache, sweating, palpitations, severe or labile hypertension, tremor, anxiety, hyperglycemia	Plasma free or urinary fractionated metanephrines; alpha blockade before beta blockade and surgery
Primary aldosteronism	Autonomous aldosterone from adrenal adenoma or bilateral adrenal hyperplasia	Hypertension, hypokalemia, metabolic alkalosis, weakness, polyuria; edema is usually absent	Aldosterone-renin ratio; confirmatory testing and subtype localization when surgery is considered
Adrenal insufficiency	Inadequate cortisol; primary disease also causes aldosterone deficiency	Fatigue, weight loss, nausea, abdominal pain, hypotension, hyponatremia; hyperkalemia and hyperpigmentation suggest primary disease	Morning cortisol with ACTH; cosyntropin stimulation; treat crisis immediately with hydrocortisone and isotonic fluids
Congenital adrenal hyperplasia	Enzyme block, most commonly 21-hydroxylase deficiency	Low cortisol, low aldosterone in salt-wasting form, androgen excess, virilization, neonatal shock	High 17-hydroxyprogesterone; replace glucocorticoid and mineralocorticoid as needed

A practical first branch is “excess versus deficiency.” Excess cortisol, aldosterone, and catecholamines generally produce hypertension; adrenal insufficiency produces hypotension unless another process coexists. Hypokalemia narrows the field toward aldosterone excess, ectopic ACTH/mineralocorticoid effects, or some forms of congenital adrenal hyperplasia. Hyperkalemia narrows the field toward primary adrenal insufficiency, hypoaldosteronism, kidney failure, or medications that block the RAAS axis.

Exam trap

Do not diagnose pheochromocytoma from panic symptoms alone. The key pattern is adrenergic spells plus objective hypertension or labile blood pressure, with biochemical confirmation before imaging. Do not give a beta-blocker before alpha blockade in a catecholamine-secreting tumor because unopposed alpha vasoconstriction can worsen hypertension.

Cushing syndrome and cortisol excess

Cushing syndrome is the clinical state of chronic glucocorticoid excess. The most common cause overall is iatrogenic exposure to prednisone, dexamethasone, or other glucocorticoids. Endogenous disease is divided into ACTH-dependent and ACTH-independent forms. Cushing disease refers specifically to an ACTH-secreting pituitary adenoma. Ectopic ACTH is often produced by neuroendocrine tumors such as small cell lung carcinoma or bronchial carcinoid. ACTH-independent Cushing syndrome usually comes from an adrenal adenoma, carcinoma, or bilateral adrenal hyperplasia.

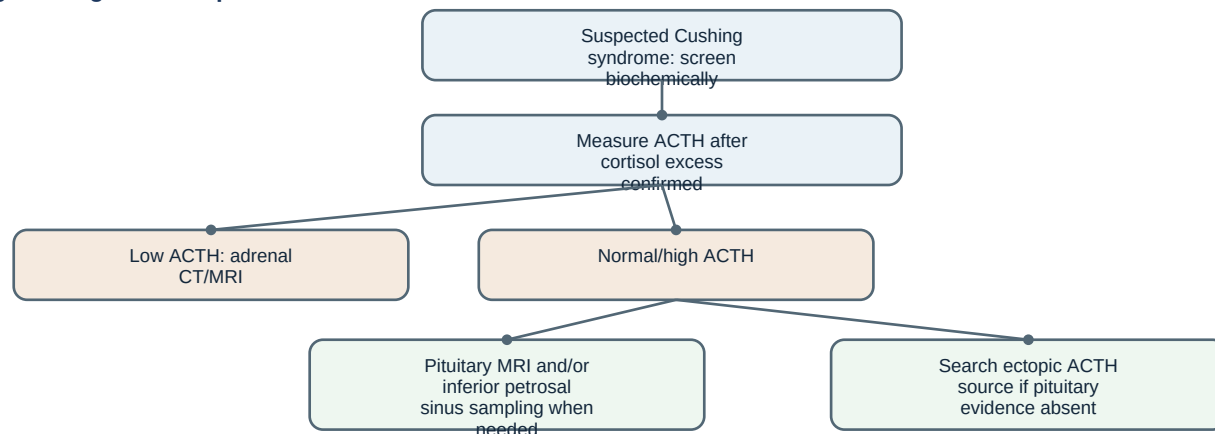
Cortisol has catabolic, immunosuppressive, and anti-insulin effects. Muscle protein breakdown produces proximal weakness. Skin and connective tissue breakdown produce wide purple striae, bruising, poor wound healing, and thin skin. Hepatic gluconeogenesis and insulin resistance produce hyperglycemia. Mineralocorticoid receptor activity and vascular sensitization produce hypertension. Cortisol also suppresses bone formation and increases fracture risk. These mechanisms explain why the “syndrome” can look like metabolic disease unless the clinician recognizes the connective tissue, muscle, menstrual, infectious, and psychiatric clues.

Finding	Interpretation
Low-dose dexamethasone fails to suppress cortisol	Supports endogenous cortisol excess after appropriate screening context
ACTH low	ACTH-independent adrenal source is likely
ACTH normal/high	ACTH-dependent source: pituitary Cushing disease or ectopic ACTH
High-dose dexamethasone suppresses cortisol by >50%	Favors pituitary Cushing disease, not definitive alone
CRH stimulation increases ACTH/cortisol	Favors pituitary source
Very high ACTH, severe hypokalemia, rapid onset	Raises concern for ectopic ACTH

Initial screening should be done only when pretest probability is meaningful, because obesity, depression, alcohol use disorder, acute illness, and poorly controlled diabetes can produce pseudo-Cushing physiology. Common screening options include

late-night salivary cortisol, 24-hour urine free cortisol, and the 1-mg overnight dexamethasone suppression test. Once hypercortisolism is established, ACTH partitions the differential. Imaging comes after biochemical classification so that incidental pituitary or adrenal lesions do not mislead the workup.

Original diagnostic map: cortisol excess



Pheochromocytoma and paraganglioma

Pheochromocytomas arise from adrenal medullary chromaffin tissue; paragangliomas arise from extra-adrenal sympathetic or parasympathetic paraganglia. The disease matters because episodes can cause hypertensive emergency, myocardial injury, stroke, arrhythmia, or collapse during anesthesia. The classic triad is episodic headache, sweating, and palpitations, but the phenotype is variable. Sustained hypertension, orthostatic hypotension from volume contraction, tremor, pallor, anxiety, hyperglycemia, and weight loss may also appear.

The biochemical screen is measurement of plasma free metanephrines or urinary fractionated metanephrines. Metanephrines are useful because catecholamines are metabolized continuously inside tumor cells, so levels can remain abnormal even between spells. Imaging is generally performed after biochemical evidence is present. Treatment is surgical resection, but preoperative preparation is the safety-critical step: alpha-adrenergic blockade is established first, volume is expanded, and beta-blockade is added only after adequate alpha blockade if tachyarrhythmias persist.

Step	High-yield point
Biochemical test	Plasma free metanephrines or urinary fractionated metanephrines
Localization	CT/MRI after biochemical confirmation; functional imaging in selected hereditary, metastatic, or extra-adrenal disease
Pre-op sequence	Alpha blockade first, then beta blockade if needed; hydrate and liberalize salt when appropriate
Dangerous error	Beta-blocker before alpha blockade can intensify vasoconstriction
Genetic clue	Young age, bilateral/multifocal tumors, paraganglioma, metastatic disease, or family history

Rule-of-10s caveat
Older teaching used a “10%” rule for malignancy, bilaterality, extra-adrenal location, and familial disease. Modern practice recognizes that hereditary disease is more common than previously thought, so genetic risk assessment is important, especially with young, bilateral, multifocal, paraganglioma, or metastatic presentations.

Primary aldosteronism

Primary aldosteronism is autonomous mineralocorticoid excess. Aldosterone increases sodium reabsorption through epithelial sodium channels in the cortical collecting duct and increases potassium and hydrogen ion secretion. The result is hypertension from extracellular volume expansion, hypokalemia, and metabolic alkalosis. Edema is usually absent because “aldosterone escape” increases natriuresis enough to limit overt swelling, but cardiovascular and renal risk remain increased.

The screening pattern is high aldosterone with suppressed renin. The aldosterone-renin ratio is most useful when interpreted with medication effects, potassium status, posture, sodium intake, and renal function in mind. Hypokalemia itself can suppress aldosterone, so potassium should be corrected before testing when possible. After biochemical screening, confirmatory suppression testing and subtype evaluation help decide therapy. Unilateral aldosterone-producing adenoma can be cured or improved with adrenalectomy; bilateral hyperplasia is treated medically with mineralocorticoid receptor antagonists such as spironolactone or eplerenone.

Component	Practical anchor
When to suspect	Resistant HTN, HTN with hypokalemia, adrenal incidentaloma with HTN, early-onset HTN, family history of early stroke/PA, HTN with sleep apnea or atrial fibrillation
Screen	Plasma aldosterone concentration and renin; elevated aldosterone-renin ratio with suppressed renin
Confirm	Saline infusion, oral sodium loading, captopril challenge, or fludrocortisone suppression depending on setting
Subtype	Adrenal CT plus adrenal venous sampling in many surgical candidates; selected young patients with clear unilateral lesions may differ by guideline context
Treat	Unilateral adenoma: adrenalectomy. Bilateral disease or nonsurgical candidate: mineralocorticoid receptor antagonist

Adrenal insufficiency and adrenal crisis

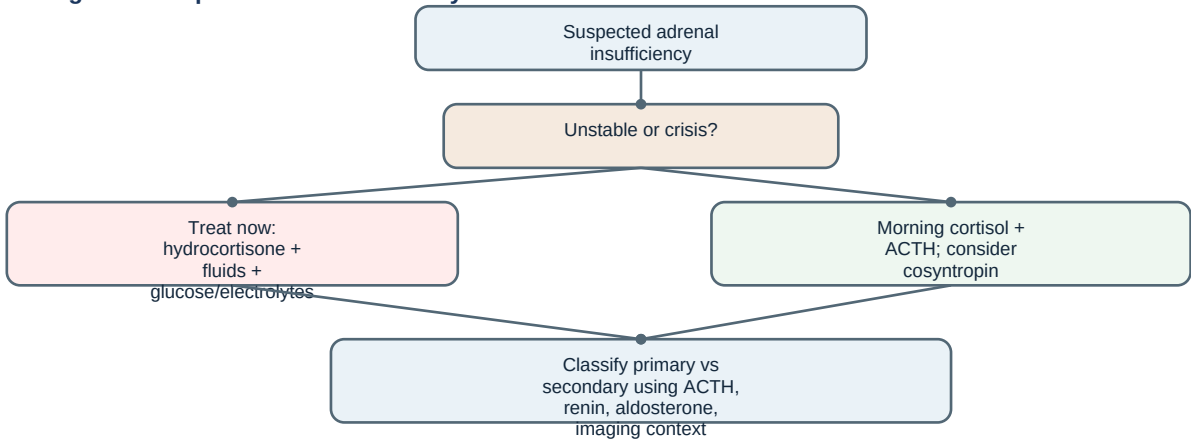
Adrenal insufficiency is inadequate cortisol effect for physiologic needs. Primary adrenal insufficiency is adrenal gland failure, so cortisol and aldosterone are both deficient. Secondary adrenal insufficiency is low ACTH from pituitary or chronic exogenous glucocorticoid suppression; aldosterone is usually preserved because the renin-angiotensin system controls the zona glomerulosa. Tertiary disease reflects hypothalamic CRH suppression or dysfunction.

Symptoms are often nonspecific until stress unmasks the disorder: fatigue, anorexia, nausea, vomiting, vague abdominal pain, weight loss, orthostatic symptoms, hyponatremia, and hypoglycemia. Primary disease adds hyperkalemia, salt craving, volume depletion, high renin, low aldosterone, and hyperpigmentation from high ACTH/POMC-derived peptides. Secondary disease lacks hyperpigmentation and usually lacks hyperkalemia. Adrenal crisis should be treated before the workup is complete if the patient is unstable.

State	Interpretation
Primary adrenal insufficiency	Low cortisol, high ACTH, low aldosterone, high renin; hyperkalemia and hyperpigmentation may occur
Secondary adrenal insufficiency	Low cortisol, low/inappropriately normal ACTH; aldosterone usually normal; no hyperpigmentation
Cosyntropin response	Adequate cortisol rise argues against primary adrenal failure; early secondary disease may have a misleading response before adrenal atrophy develops
Adrenal crisis	Hypotension/shock, vomiting, abdominal pain, fever, hypoglycemia, hyponatremia, hyperkalemia; treat immediately

Adrenal crisis treatment
Draw cortisol and ACTH if doing so does not delay care. Give stress-dose IV hydrocortisone, isotonic saline with dextrose as needed, correct hypoglycemia and hyperkalemia, and search for precipitating infection, surgery, trauma, or abrupt steroid withdrawal.

Original diagnostic map: adrenal insufficiency



Congenital adrenal hyperplasia

Congenital adrenal hyperplasia is a group of inherited adrenal steroid synthesis defects. The classic board pattern is 21-hydroxylase deficiency: decreased cortisol, decreased aldosterone in salt-wasting disease, increased ACTH, adrenal hyperplasia, and shunting of steroid precursors into androgen production. Female newborns can have virilized external genitalia with internal female reproductive organs; male newborns may have no obvious genital abnormality but can present with life-threatening salt wasting in the first weeks of life.

The biochemical anchor is elevated 17-hydroxyprogesterone. Severe salt-wasting disease causes vomiting, dehydration, shock, hyponatremia, hyperkalemia, and hypoglycemia. Long-term management replaces glucocorticoid to suppress excess ACTH and androgen production, replaces mineralocorticoid/salt when needed, and tracks growth, puberty, fertility, and overtreatment complications.

Enzyme pattern	Clinical signature
21-hydroxylase deficiency	Low cortisol, low aldosterone in salt-wasting form, high androgens, high 17-hydroxyprogesterone, hyponatremia/hyperkalemia

Enzyme pattern	Clinical signature
11-beta-hydroxylase deficiency	Low cortisol, high androgens, mineralocorticoid precursor excess; hypertension and hypokalemia
17-alpha-hydroxylase deficiency	Low cortisol and sex steroids, mineralocorticoid excess; hypertension, hypokalemia, sexual infantilism/undervirilization

Integrated diagnostic approach

Adrenal questions become easier if the first lab cluster is interpreted physiologically rather than memorized as isolated facts. Hypertension plus hypokalemia points to mineralocorticoid effect: primary aldosteronism if renin is suppressed, renovascular disease if renin is high, cortisol excess if Cushing features coexist, or congenital enzyme defects if age/sex development clues are present. Episodic hypertension plus adrenergic spells points toward catecholamine excess. Hypotension plus hyponatremia points toward adrenal insufficiency, severe hypothyroidism, volume depletion, SIADH, or medications; hyperkalemia strongly supports primary adrenal or renal/mineralocorticoid pathway disease.

Pattern	Prioritized differential
HTN + hypokalemic metabolic alkalosis	Primary aldosteronism, renovascular HTN, Cushing/ectopic ACTH, licorice, Liddle syndrome, 11-beta/17-alpha hydroxylase deficiency
HTN + episodic headache/sweats/palpitations	Pheochromocytoma/paraganglioma until proven otherwise
Obesity + diabetes + proximal weakness + bruising/striae	Cushing syndrome more likely than simple metabolic syndrome
Hypotension + hyponatremia + hyperkalemia	Primary adrenal insufficiency, hypoadosteronism, kidney failure, RAAS-blocking drugs
Hyponatremia + low cortisol without hyperkalemia	Secondary adrenal insufficiency can mimic SIADH
Virilization + salt wasting neonate	Classic CAH, especially 21-hydroxylase deficiency

Medication history is part of the diagnostic test. Exogenous glucocorticoids can cause Cushingoid features while suppressing ACTH and endogenous cortisol. Abrupt withdrawal after chronic use can cause secondary adrenal insufficiency. Diuretics can confound aldosterone-renin and potassium interpretation. Oral contraceptives, antidepressants, stimulants, decongestants, and acute illness can influence endocrine testing or mimic symptoms. A clinically useful endocrine note always records current steroids, prior steroid injections, antihypertensives, diuretics, supplements, and recent illness.

Initial management by syndrome

Syndrome	Management priorities
Cushing syndrome	Treat hypertension, diabetes, infection risk, VTE risk, psychiatric symptoms, osteoporosis; remove causative tumor when possible; taper iatrogenic steroids safely
Pheochromocytoma	Avoid triggers, establish alpha blockade, volume expansion, then beta blockade if needed; surgical resection by experienced team
Primary aldosteronism	Control BP and potassium; adrenalectomy for unilateral disease; spironolactone/eplerenone for bilateral disease or nonsurgical patients
Adrenal crisis	Immediate hydrocortisone, isotonic saline, dextrose as needed, electrolyte correction, treat trigger
Chronic primary adrenal insufficiency	Physiologic glucocorticoid replacement plus fludrocortisone if aldosterone deficient; stress-dose education and emergency steroid plan
CAH	Glucocorticoid replacement to suppress ACTH; mineralocorticoid/salt when deficient; multidisciplinary pediatric/endocrine/urology care as needed

High-yield exam clues and common pitfalls

- **Wide purple striae plus proximal weakness** is a stronger Cushing clue than obesity alone.
- **ACTH-dependent Cushing** can cause hyperpigmentation; adrenal tumors suppress ACTH and usually do not.
- **Pheochromocytoma** should be biochemically confirmed before imaging because adrenal incidentalomas are common.
- **Alpha before beta** is the perioperative rule for catecholamine-secreting tumors.
- **Primary aldosteronism can be normokalemic.** Do not require hypokalemia before screening a high-risk hypertensive patient.
- **Primary adrenal insufficiency** causes hyperkalemia; secondary adrenal insufficiency usually does not.
- **Adrenal crisis** is a treatment-first diagnosis. Do not delay hydrocortisone in shock.
- **CAH** is not one disease; enzyme location predicts salt wasting versus hypertension and androgen excess versus deficiency.

Diagnostic trap

A "normal" random cortisol is not always reassuring in a critically ill hypotensive patient. In stress, cortisol should be appropriately elevated. A value that is technically in the reference range may be inadequate for the physiologic context.

Practice questions

1. A 43-year-old woman has diabetes, new hypertension, proximal leg weakness, easy bruising, and wide purple abdominal striae. Morning cortisol is high after a 1-mg dexamethasone test. ACTH is low. What is the next localization step?

Answer: Adrenal imaging is appropriate because low ACTH indicates ACTH-independent cortisol production, commonly adrenal adenoma/carcinoma or adrenal hyperplasia.

2. A 36-year-old man has episodic pounding headaches, sweating, palpitations, and BP 220/120 during attacks. CT shows an adrenal mass. What should happen before beta-blockade or surgery?

Answer: Biochemical confirmation with plasma or urinary metanephrines and preoperative alpha blockade. Beta-blockade should not precede alpha blockade.

3. A patient with resistant hypertension has K 2.9 mEq/L and bicarbonate 34 mEq/L. Plasma renin is suppressed and aldosterone is elevated. What disorder is most likely?

Answer: Primary aldosteronism. The pattern is autonomous aldosterone causing sodium retention, potassium wasting, and hydrogen ion loss.

4. A 29-year-old with autoimmune thyroid disease has fatigue, weight loss, nausea, orthostasis, Na 126, K 5.9, and diffuse hyperpigmentation. ACTH is high and cortisol is low. What therapy is needed chronically?

Answer: Primary adrenal insufficiency requires glucocorticoid replacement and mineralocorticoid replacement when aldosterone deficient, commonly hydrocortisone plus fludrocortisone.

5. A newborn has vomiting, dehydration, hyponatremia, hyperkalemia, hypoglycemia, and ambiguous external genitalia with internal female reproductive structures. What enzyme defect is most likely?

Answer: Classic salt-wasting 21-hydroxylase deficiency, with elevated 17-hydroxyprogesterone and androgen excess.

Graph-RAG extraction map

Node	Class	Key attributes	Useful edges
Cushing syndrome	Disease	cortisol excess; ACTH-dependent/independent	causes -> diabetes, HTN, osteoporosis; diagnosed_by -> dex suppression/urine cortisol/salivary cortisol; localized_by -> ACTH
Pheochromocytoma	Tumor/syndrome	catecholamine excess	presents_with -> headache/sweating/palpitations; diagnosed_by -> metanephrines; treated_by -> alpha blockade then adrenalectomy
Primary aldosteronism	Mineralocorticoid disorder	high aldosterone + low renin	causes -> HTN/hypokalemia/metabolic alkalosis; treated_by -> adrenalectomy or MRA
Adrenal insufficiency	Hormone deficiency	low cortisol; primary adds low aldosterone	causes -> hypotension/hyponatremia; primary_causes -> hyperkalemia/hyperpigmentation; treated_by -> hydrocortisone

Node	Class	Key attributes	Useful edges
Adrenal crisis	Emergency	acute cortisol deficiency under stress	managed_by -> stress-dose steroids + fluids + dextrose + trigger control
CAH	Inherited enzyme disorder	steroid synthesis block	21-hydroxylase_deficiency -> high 17-OHP + salt wasting + virilization

Source and guideline notes

Coverage signals: uploaded endocrine adrenal source section, pages covering Cushing syndrome, pheochromocytoma, primary hyperaldosteronism, adrenal insufficiency, and congenital adrenal hyperplasia. Current clinical cross-checks: Endocrine Society adrenal guideline resources; Endocrine Society guideline resources for Cushing syndrome treatment, primary adrenal insufficiency, primary aldosteronism, pheochromocytoma/paraganglioma, and congenital adrenal hyperplasia.

This chapter is designed for medical education and graph-RAG extraction. It is not a substitute for local protocols, specialist consultation, or individualized patient care.